

绝对值与相反数的计算

一、相反数

$$(1) +(-1) = \underline{\hspace{2cm}}$$

$$-(-1) = \underline{\hspace{2cm}} ;$$

$$(2) -\left(-\frac{3}{2}\right) = \underline{\hspace{2cm}}$$

$$-(+6) = \underline{\hspace{2cm}} ;$$

$$(3) -(-1.3) = \underline{\hspace{2cm}}$$

$$-(+2.5) = \underline{\hspace{2cm}} ;$$

$$(4) -\left(+\frac{3}{4}\right) = \underline{\hspace{2cm}}$$

$$+\left(+\frac{1}{3}\right) = \underline{\hspace{2cm}} ;$$

$$(5) -(-15) = \underline{\hspace{2cm}} ;$$

$$-(+6.9) = \underline{\hspace{2cm}} ;$$

$$(6) -[-(-1.3)] = \underline{\hspace{2cm}}$$

$$+[-(+2)] = \underline{\hspace{2cm}} ;$$

$$(7) -[+(+3)] = \underline{\hspace{2cm}}$$

$$(3)-(-1.3) = \underline{\hspace{2cm}} ;$$

$$(8) +\left[-\left(-\frac{1}{4}\right)\right] = \underline{\hspace{2cm}}$$

$$-\left[+\left(-3\frac{2}{3}\right)\right] = \underline{\hspace{2cm}} ;$$

$$(9) -[+(-0.75)] = \underline{\hspace{2cm}}$$

$$+[-(-68)] = \underline{\hspace{2cm}} ;$$

$$(10) -[-(+3.8)] = \underline{\hspace{2cm}}$$

$$-[-(-8)] = \underline{\hspace{2cm}} ;$$

$$(11) +\left[+\left(-\frac{15}{16}\right)\right] = \underline{\hspace{2cm}}$$

$$-\left[+\left(+8\frac{3}{8}\right)\right] = \underline{\hspace{2cm}} ;$$

$$(12) -\{-[+(-0.3)]\} = \underline{\hspace{2cm}}$$

$$-\{+[+(-6.8)]\} = \underline{\hspace{2cm}} ;$$

$$(13) +\{-[-(+9)]\} = \underline{\hspace{2cm}}$$

$$+\{-[-(-8.4)]\} = \underline{\hspace{2cm}} ;$$

$$(14) +\{+[+(-24)]\} = \underline{\hspace{2cm}}$$

$$-\left\{-\left[-\left(-\frac{13}{18}\right)\right]\right\} = \underline{\hspace{2cm}} ;$$

$$(15) +\{+[-(-0)]\} = \underline{\hspace{2cm}}$$

$$-\{+[-(-0)]\} = \underline{\hspace{2cm}} .$$

二、绝对值

(1) $|-1| = \underline{\hspace{2cm}}$

(2) $|-8.5| = \underline{\hspace{2cm}}$

(3) $\left|\frac{1}{4}\right| = \underline{\hspace{2cm}}$

(4) $| -(-1.5) | = \underline{\hspace{2cm}}$

(5) $\left| -\left(+\frac{2}{3}\right) \right| = \underline{\hspace{2cm}}$

(6) $-|-2007| = \underline{\hspace{2cm}}$

(7) $+\left| -\frac{2}{5} \right| = \underline{\hspace{2cm}}$

(8) $-|-(+3)| = \underline{\hspace{2cm}}$

(9) $+|+(+2.9)| = \underline{\hspace{2cm}}$

(10) $-| -(-16) | = \underline{\hspace{2cm}}$

(11) $|a| = \underline{\hspace{2cm}} (a > 0)$

(12) $|-a| = \underline{\hspace{2cm}} (a > 0)$

(13) $-|m| = \underline{\hspace{2cm}} (m > 0)$

(14) $+|-m| = \underline{\hspace{2cm}} (m < 0)$

(15) $-|+(-a)| = \underline{\hspace{2cm}} (a < 0)$

$|+3| = \underline{\hspace{2cm}} ;$

$\left| -\frac{3}{4} \right| = \underline{\hspace{2cm}} ;$

$\left| -\frac{1}{2012} \right| = \underline{\hspace{2cm}} ;$

$|+(-11)| = \underline{\hspace{2cm}} ;$

$\left| +\left(+\frac{2}{5}\right) \right| = \underline{\hspace{2cm}} ;$

$-|+2018| = \underline{\hspace{2cm}} ;$

$+|+5.1| = \underline{\hspace{2cm}} ;$

$+\left| -\left(+\frac{5}{7}\right) \right| = \underline{\hspace{2cm}} ;$

$-|-(+3.2)| = \underline{\hspace{2cm}} ;$

$+|+(-25)| = \underline{\hspace{2cm}} ;$

$|a| = \underline{\hspace{2cm}} (a < 0) ;$

$|-a| = \underline{\hspace{2cm}} (a < 0) ;$

$-|-m| = \underline{\hspace{2cm}} (m < 0) ;$

$-|+a| = \underline{\hspace{2cm}} (a > 0) ;$

$-|-(+a)| = \underline{\hspace{2cm}} (a > 0) .$

► 添括号

1. 在下列各式的括号内填上恰当的项：

- (1) $-a+b-c+d=-a+(\quad)$;
- (2) $-a+b-c+d=-(\quad)+d$;
- (3) $-a+b-c+d=-a+b-(\quad)$;
- (4) $-a+b-c+d=-(\quad)$.

2. 在下列各式的括号里，填上适当的项：

- (1) $(a+b+c)(a-b+c)=[a+(\quad)][a-(\quad)]$;
- (2) $(a-b+c)(a+b-c)=[a-(\quad)][a+(\quad)]$;
- (3) $(-a+b+c)(a+b-c)=[b-(\quad)][b+(\quad)]$;
- (4) $(a+b-c)(a-b-c)=[a+(\quad)][a-(\quad)]$;
- (5) $(a+b-c-d)(a-b+c-d)=[(a\quad)+(b\quad)][(a\quad)-(b\quad)]$.

3. 在括号内填上适当的项：

- (1) $x^2-y^2+8y-4=x^2-(\quad)$;
- (2) $(-a+2b+3c)(a+2b-3c)=[2b-(\quad)][2b+(a-3c)]$;
- (3) $ax-bx-ay+by=(ax-bx)-(\quad)$;
- (4) $5x+3x^2-4y^2=5x-(\quad)$;
- (5) $-3p+3q-1=3q-(\quad)$;
- (6) $(-2a+3b+5c)(2a+3b-5c)=[3b-(\quad)][3b+(\quad)]$.

4. 在等式的括号内填上适当的项：

- (1) $1-a^2+2ab-b^2=1-(\quad)$;
- (2) $a-(b-c+d)=a-d+(\quad)$;
- (3) $a^2-ab+b^2=a^2-(\quad)$;
- (4) $2x-3y+3z=2x-3(\quad)$;
- (5) $2a-3b+4c=2a-(\quad)$;
- (6) $2x^2y^3-3xy+1=2x^2y^3-(\quad)$.

5. 把多项式 $-x^3-6x^2y+12xy^2-8y^3+1$ 写成两个整式的和，使其中一个不含字母 x ，结果是

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► 一次多项式合并

1. 计算:

$$\begin{array}{ll} (1) 12x - 20x = \underline{\hspace{2cm}}; & (2) -a - a - 2a = \underline{\hspace{2cm}}; \\ (3) \frac{1}{2}y - 3y + 2y = \underline{\hspace{2cm}}; & (4) \frac{2}{3}a - \frac{5}{3}a + 8a = \underline{\hspace{2cm}}. \end{array}$$

2. 合并同类项:

$$\begin{array}{ll} (1) 3x + 2y - 5x - y = \underline{\hspace{2cm}}; & (2) 3b + 5a + 2a - 4b = \underline{\hspace{2cm}}; \\ (3) -3a + 2b - 7a + 3b + 3 = \underline{\hspace{2cm}}; & (4) 5a - 2b + 3b - 4a - 1 = \underline{\hspace{2cm}}; \\ (5) x - f + 5x - 4f = \underline{\hspace{2cm}}; & (6) -12x + 6y + 10x - y = \underline{\hspace{2cm}}; \\ (7) -\frac{3}{8}x + \frac{2}{15}y - \frac{5}{8}x - \frac{7}{15}y = \underline{\hspace{2cm}}; & (8) \frac{2}{3}x - \frac{6}{11}y - \frac{1}{4} - \frac{3}{4} + \frac{4}{3}x - \frac{5}{11}y = \underline{\hspace{2cm}}. \end{array}$$

3. 将下列各式去括号, 并合并同类项:

$$\begin{array}{l} (1) 3x - 2y - [4x - 3(x - y)] = \underline{\hspace{2cm}}; \\ (2) 5(m + 3) - (m - 4) + 2(m + 1) = \underline{\hspace{2cm}}; \\ (3) 2a - (5a - 3b) + 3(2a - b) = \underline{\hspace{2cm}}; \\ (4) 3(x - 3y) - 2(y - 2x) - x = \underline{\hspace{2cm}}; \\ (5) 2(2a - 3b) + 3(2b - 3a) = \underline{\hspace{2cm}}; \\ (6) 3a - (a - 3b) - (a + 2b) - 2(a - b) = \underline{\hspace{2cm}}; \\ (7) 5(2x - 7y) - 3(4x - 10y) = \underline{\hspace{2cm}}; \\ (8) \left(\frac{7}{15}y - \frac{1}{3}z\right) - 2\left(\frac{3}{15}y - \frac{5}{3}z\right) = \underline{\hspace{2cm}}; \\ (9) -\frac{1}{8}x - \left(\frac{3}{8}x - 1\right) = \underline{\hspace{2cm}}; \\ (10) 2x - [2(x + 3y) - 3(x - 2y)] = \underline{\hspace{2cm}}; \end{array}$$

4. 合并同类项:

$$\begin{array}{l} (1) a + b - 4(a + b) = \underline{\hspace{2cm}}; \\ (2) 7(a - b) - 3(a - b) - 2(a - b) = \underline{\hspace{2cm}}; \\ (3) 2(a + b) - 5(a + b) + a + b = \underline{\hspace{2cm}}; \\ (4) 3(a - b) + 2(a - b) - 11(a - b) = \underline{\hspace{2cm}}; \\ (5) 3a + 3b - a - b + \frac{2}{3}a + \frac{2}{3}b - \frac{1}{2}a - \frac{1}{2}b = \underline{\hspace{2cm}}; \\ (6) 4(3a - b) - (3a - b) + 5(b - 3a) + 2(b - 3a) = \underline{\hspace{2cm}}; \\ (7) a - 2a + 3a - 4a + 5a - 6a + \dots 99a - 100a = \underline{\hspace{2cm}}; \\ (8) a + 2a + 3a + 4a + 5a + 6a + \dots 99a + 100a = \underline{\hspace{2cm}}. \end{array}$$

一、整式加减运算

1. $(5a^2 - 3b^2) + (a^2 + b^2) - (5a^2 + 3b^2)$

2. $3(2x^2 - y^2) - 2(3y^2 - 2x^2)$

3. $(8xy - x^2 + y^2) - (x^2 - y^2 + 8xy)$

4. $2(2a - 3b) + 3(2b - 3a)$

5. $3x^2 - [7x - (4x - 3) - 2x^2]$

6. $(5a^2 - 2ab) - 2(3a^2 + 4ab - \frac{1}{2}b^2)$

7. $-4(pq + pr) + (4pq + pr)$

8. $(x^3 - 2y^3 - 3x^2y) - (3x^3 - 3y^3 - 7x^2y)$

9. $5a^2b - [2ab^2 - 3(ab^2 - a^2b)]$

10. $3(-ab + 2a) - (3a - b) + 3ab$

11. $2(x^2 - xy) - 3(2x^2 - 3xy) - 2[x^2 - (2x^2 - xy + y^2)]$

12. $\frac{1}{2}x - 2(x - \frac{1}{3}y^2) + (-\frac{3}{2}x + \frac{1}{3}y^2)$

13. 已知 $A = 2x^2 + 3xy - 2x - 1$, $B = -x^2 + xy - 1$, 计算 $3A + 6B$

14. 已知 $A = 3a^2 - 2a + 1$, $B = 5a^2 - 3a + 2$, 计算 $2A - 3B$

15. $-3(2s - 5) + 6s$

16. $1 - 3(2ab + a) + [1 - 2(2a - 3ab)]$

17. $(4a^2b - 3ab) + (-5a^2b + 2ab)$

18. $(6m^2 - 4m - 3) + (2m^2 - 4m + 1)$

19. $(8xy - x^2 + y^2) - 4(x^2 - y^2 + 2xy - 3)$

20. $5(x + y) - 4(3x - 2y) + 3(2x - 3y)$

绝对值与相反数的计算

一、相反数

$$(1) +(-1) = \underline{-1}$$

$$-(-1) = \underline{1};$$

$$(2) -\left(-\frac{3}{2}\right) = \underline{\frac{3}{2}}$$

$$-(+6) = \underline{-6};$$

$$(3) -(-1.3) = \underline{1.3}$$

$$-(+2.5) = \underline{-2.5};$$

$$(4) -\left(+\frac{3}{4}\right) = \underline{-\frac{3}{4}}$$

$$+\left(+\frac{1}{3}\right) = \underline{\frac{1}{3}};$$

$$(5) -(-15) = \underline{15}$$

$$-(+6.9) = \underline{-6.9};$$

$$(6) -[-(-1.3)] = \underline{-1.3}$$

$$+[-(+2)] = \underline{-2};$$

$$(7) -[+(+3)] = \underline{-3}$$

$$+[+(-8.1)] = \underline{-8.1};$$

$$(8) +\left[-\left(-\frac{1}{4}\right)\right] = \underline{\frac{1}{4}}$$

$$-\left[+\left(-3\frac{2}{3}\right)\right] = \underline{3\frac{2}{3}};$$

$$(9) -[+(-0.75)] = \underline{0.75}$$

$$+[-(-68)] = \underline{68};$$

$$(10) -[-(+3.8)] = \underline{3.8}$$

$$-[-(-8)] = \underline{8};$$

$$(11) +\left[+\left(-\frac{15}{16}\right)\right] = \underline{-\frac{15}{16}}$$

$$-\left[+\left(+8\frac{3}{8}\right)\right] = \underline{-8\frac{3}{8}};$$

$$(12) -\{-[-(+0.3)]\} = \underline{-0.3}$$

$$-\{+[-(-6.8)]\} = \underline{6.8};$$

$$(13) +\{-[-(+9)]\} = \underline{9}$$

$$+\{-[-(-8.4)]\} = \underline{8.4};$$

$$(14) +\{+[-(-24)]\} = \underline{-24}$$

$$-\left\{-\left[-\left(-\frac{13}{18}\right)\right]\right\} = \underline{-\frac{13}{18}};$$

$$(15) +\{+[-(-0)]\} = \underline{0}$$

$$-\{+[-(-0)]\} = \underline{0}.$$

二、绝对值

$$(1) |-1| = \underline{1}$$

$$(2) |-8.5| = \underline{8.5}$$

$$(3) \left| \frac{1}{4} \right| = \underline{\frac{1}{4}}$$

$$(4) |-(-1.5)| = \underline{1.5}$$

$$(5) \left| -\left(+\frac{2}{3} \right) \right| = \underline{\frac{2}{3}}$$

$$(6) -|-2007| = \underline{-2007}$$

$$(7) +\left| -\frac{2}{5} \right| = \underline{\frac{2}{5}}$$

$$(8) -|-(+3)| = \underline{-3}$$

$$(9) +|+(+2.9)| = \underline{2.9}$$

$$(10) -|-(-16)| = \underline{-16}$$

$$(11) |a| = \underline{a} \quad (a > 0)$$

$$(12) |-a| = \underline{a} \quad (a > 0)$$

$$(13) -|m| = \underline{-m} \quad (m > 0)$$

$$(14) +|-m| = \underline{-m} \quad (m < 0)$$

$$(15) -|+(-a)| = \underline{a} \quad (a < 0)$$

$$|+3| = \underline{3};$$

$$\left| -\frac{3}{4} \right| = \underline{\frac{3}{4}};$$

$$\left| -\frac{1}{2012} \right| = \underline{\frac{1}{2012}};$$

$$|+(-11)| = \underline{11};$$

$$\left| +\left(+\frac{2}{5} \right) \right| = \underline{\frac{2}{5}};$$

$$-|+2018| = \underline{-2018};$$

$$+|+5.1| = \underline{5.1};$$

$$+\left| -\left(+\frac{5}{7} \right) \right| = \underline{\frac{5}{7}};$$

$$-|- (+3.2)| = \underline{-3.2};$$

$$+|+(-25)| = \underline{25};$$

$$|a| = \underline{-a} \quad (a < 0);$$

$$|-a| = \underline{-a} \quad (a < 0);$$

$$-|-m| = \underline{m} \quad (m < 0);$$

$$-|+a| = \underline{-a} \quad (a > 0);$$

$$-|- (+a)| = \underline{-a} \quad (a > 0).$$

合并同类项

一、带括号、带系数

去括号

1. 去括号：

$$(1) -(a-2b-c) = \underline{-a+2b+c};$$

$$(2) -(-7x+5) = \underline{7x-5};$$

$$(3) -(a+2b-3c) = \underline{-a-2b+3c};$$

$$(4) a-(-b+c-d) = \underline{a+b-c+d}.$$

2. 化简：

$$(1) -2(m-n) = \underline{-2m+2n};$$

$$(2) -2a-(-2a-1) = \underline{1};$$

$$(3) -[-(+5)] = \underline{5};$$

$$(4) -[-(-a+b)-c] = \underline{-a+b+c}.$$

$$3. \text{ 已知 } -a=5, \text{ 则 } -[+(-a)] = \underline{-5}.$$

$$4. \text{ 把多项式: } x^5 - (-4x^4y + 5xy^4) - 6(-x^3y^2 + x^2y^3) + (-3y^5) \text{ 去括号后按字母 } x \text{ 的降幂排列为 } \underline{x^5 + 4x^4y + 6x^3y^2 - 6x^2y^3 - 5xy^4 - 3y^5}$$

5. 去掉下列各式中的括号：

$$(1) (x+y)-z = \underline{x+y-z};$$

$$(2) x-(y+z) = \underline{x-y-z};$$

$$(3) -1-2(x-y) = \underline{-1-2x+2y};$$

$$(4) 2(a-b)-3(x+y) = \underline{2a-2b-3x-3y};$$

$$(5) x+3(-2y+z) = \underline{x-6y+3z};$$

$$(6) x-5(2y-3z) = \underline{x-10y+15z};$$

$$(7) 3a^2-2(a-b-5c) = \underline{3a^2-2a+2b+10c};$$

$$(8) -a^3-(-3a^2+2a-1) = \underline{-a^3+3a^2-2a+1};$$

$$(9) (x-y)-(m+n) = \underline{x-y-m-n};$$

$$(10) a-\frac{3}{5}(5b-10) = \underline{a-3b+b}.$$

6. 把下列各式的括号去掉：

$$(1) (a-b)+(-c-d) = \underline{a-b-c-d};$$

$$(2) -(a-b)-(-c-d) = \underline{-a+b+c+d};$$

$$(3) 5x^3-[3x^2-(x-1)] = \underline{5x^3-3x^2+x+1};$$

$$(4) 2(a-b)-3(x+y) = \underline{2a-2b-3x-3y};$$

$$(5) (m-n)(p-q) = \underline{mp-mq-np+nq};$$

$$(6) -2a^2-[3a^3-(a-2)] = \underline{-2a^2-3a^3+a+2};$$

$$(7) n+2(4-2m) = \underline{n+8-4m};$$

$$(8) -(3a+b)+(a-3b) = \underline{-2a-4b};$$

$$(9) -a^3-(-3a^2+2a-1) = \underline{-a^3+3a^2-2a+1};$$

$$(10) 3a^2-2(a-b-5c) = \underline{3a^2-2a+2b+10c}$$

► 添括号

1. 在下列各式的括号内填上恰当的项：

- (1) $-a+b-c+d = -a + (\underline{b-c+d})$;
- (2) $-a+b-c+d = -(\underline{a-b+c}) + d$;
- (3) $-a+b-c+d = -a+b - (\underline{c-d})$;
- (4) $-a+b-c+d = -(\underline{a-b+c-d})$.

2. 在下列各式的括号里，填上适当的项：

- (1) $(a+b+c)(a-b+c) = [a + \underline{(b+c)}][a - \underline{(b-c)}]$;
- (2) $(a-b+c)(a+b-c) = [a - \underline{(b-c)}][a + \underline{(b-c)}]$;
- (3) $(-a+b+c)(a+b-c) = [b - \underline{(a-c)}][b + \underline{(a-c)}]$;
- (4) $(a+b-c)(a-b-c) = [a + \underline{(b-c)}][a - \underline{(b+c)}]$;
- (5) $(a+b-c-d)(a-b+c-d) = [(a - \underline{c}) + (b - \underline{d})][(a + \underline{c}) - (b + \underline{d})]$.

3. 在括号内填上适当的项：

- (1) $x^2 - y^2 + 8y - 4 = x^2 - (\underline{y^2 - 8y + 4})$;
- (2) $(-a+2b+3c)(a+2b-3c) = [2b - (\underline{a-3c})][2b + (a-3c)]$;
- (3) $ax - bx - ay + by = (ax - bx) - (\underline{ay - by})$;
- (4) $5x + 3x^2 - 4y^2 = 5x - (\underline{4y^2 - 3x^2})$;
- (5) $-3p + 3q - 1 = 3q - (\underline{3p + 1})$;
- (6) $(-2a+3b+5c)(2a+3b-5c) = [3b - (\underline{2a-5c})][3b + (\underline{2a-5c})]$.

4. 在等式的括号内填上适当的项：

- (1) $1 - a^2 + 2ab - b^2 = 1 - (\underline{a^2 - 2ab + b^2})$;
- (2) $a - (b - c + d) = a - d + (\underline{c - b})$;
- (3) $a^2 - ab + b^2 = a^2 - (\underline{ab - b^2})$;
- (4) $2x - 3y + 3z = 2x - 3(\underline{y - z})$;
- (5) $2a - 3b + 4c = 2a - (\underline{3b - 4c})$;
- (6) $2x^2y^3 - 3xy + 1 = 2x^2y^3 - (\underline{3xy - 1})$.

5. 把多项式 $-x^3 - 6x^2y + 12xy^2 - 8y^3 + 1$ 写成两个整式的和，使其中一个不含字母 x ，结果是

$$\underline{(-x^3 - 6x^2y + 12xy^2) + (1 - 8y^3)}.$$

►一次多项式合并

1. 计算:

$$(1) 12x - 20x = \underline{-8x};$$

$$(2) -a - a - 2a = \underline{-4a};$$

$$(3) \frac{1}{2}y - 3y + 2y = \underline{-\frac{1}{2}y};$$

$$(4) \frac{2}{3}a - \frac{5}{3}a + 8a = \underline{7a}.$$

2. 合并同类项:

$$(1) 3x + 2y - 5x - y = \underline{-2x + y};$$

$$(2) 3b + 5a + 2a - 4b = \underline{7a - b};$$

$$(3) -3a + 2b - 7a + 3b + 3 = \underline{-10a + 5b + 3};$$

$$(4) 5a - 2b + 3b - 4a - 1 = \underline{a + b - 1};$$

$$(5) x - f + 5x - 4f = \underline{6x - 5f};$$

$$(6) -12x + 6y + 10x - y = \underline{-2x + 5y};$$

$$(7) -\frac{3}{8}x + \frac{2}{15}y - \frac{5}{8}x - \frac{7}{15}y = \underline{-x - \frac{1}{3}y};$$

$$(8) \frac{2}{3}x - \frac{6}{11}y - \frac{1}{4} - \frac{3}{4} + \frac{4}{3}x - \frac{5}{11}y = \underline{\frac{2}{3}x - y - 1}.$$

3. 将下列各式去括号, 并合并同类项:

$$(1) 3x - 2y - [4x - 3(x - y)] = \underline{-x - 5y};$$

$$(2) 5(m + 3) - (m - 4) + 2(m + 1) = \underline{6m + 21};$$

$$(3) 2a - (5a - 3b) + 3(2a - b) = \underline{3a};$$

$$(4) 3(x - 3y) - 2(y - 2x) - x = \underline{6x - 11y};$$

$$(5) 2(2a - 3b) + 3(2b - 3a) = \underline{-5a};$$

$$(6) 3a - (a - 3b) - (a + 2b) - 2(a - b) = \underline{-a + 3b};$$

$$(7) 5(2x - 7y) - 3(4x - 10y) = \underline{-2x - 5y};$$

$$(8) \left(\frac{7}{15}y - \frac{1}{3}z\right) - 2\left(\frac{3}{15}y - \frac{5}{3}z\right) = \underline{\frac{1}{15}y + 3z};$$

$$(9) -\frac{1}{8}x - \left(\frac{3}{8}x - 1\right) = \underline{-\frac{1}{2}x + 1};$$

$$(10) 2x - [2(x + 3y) - 3(x - 2y)] = \underline{3x - 12y};$$

4. 合并同类项:

$$(1) a + b - 4(a + b) = \underline{-3a - 3b};$$

$$(2) 7(a - b) - 3(a - b) - 2(a - b) = \underline{2a - 2b};$$

$$(3) 2(a + b) - 5(a + b) + a + b = \underline{-2a - 2b};$$

$$(4) 3(a - b) + 2(a - b) - 11(a - b) = \underline{-6a + 6b};$$

$$(5) 3a + 3b - a - b + \frac{2}{3}a + \frac{2}{3}b - \frac{1}{2}a - \frac{1}{2}b = \underline{\frac{13}{6}a + \frac{13}{6}b};$$

$$(6) 4(3a - b) - (3a - b) + 5(b - 3a) + 2(b - 3a) = \underline{4b - 12a};$$

$$(7) a - 2a + 3a - 4a + 5a - 6a + \dots - 99a - 100a = \underline{-50a};$$

$$(8) a + 2a + 3a + 4a + 5a + 6a + \dots - 99a + 100a = \underline{5050a}.$$

一、整式加减答案

1. $a^2 - 5b^2$
2. $10x^2 - 9y^2$
3. $-2x^2 + 2y^2$
4. $-5a$
5. $5x^2 - 3x - 3$
6. $-a^2 - 10ab + b^2$
7. $-3pr$
8. $-2x^3 + y^3 + 4x^2y$
9. $2a^2b + ab^2$
10. $3a + b$
11. $-2x^2 + 5xy + 2y^2$
12. $-3x + y^2$
13. $15xy - 6x - 9$
14. $-9a^2 + 5a - 4$
15. 15
16. $2 - 7a$
17. $-a^2b - ab$
18. $8m^2 - 8m - 2$
19. $-5x^2 + 5y^2 + 12$
20. $-x + 4y$

二、幂的运算答案

1. 0
2. $-24x^7y^5$
3. $x - y$
4. -2
5. 45
6. 0
7. $-a^4$
8. $-100a^9$
9. $11x^6$
10. $-(x - y)^8$
11. 5400
12. 8
13. $-(m - n)^9$
14. $-16a^6b^3$
15. 243
16. $3p^6$
17. 0
18. 2^{16}
19. $x = \frac{25}{7}$
20. $2x^9$